You are given an IP address and subnet mask for the network as follows:

IP: 126.64.33.10

Subnet Mask: 255.0.0.0

a. What is the class of this IP?

b. What is the network ID for this Network?

c. What is the broadcast ID for this network?

d. What is the usable ID rage for this network?

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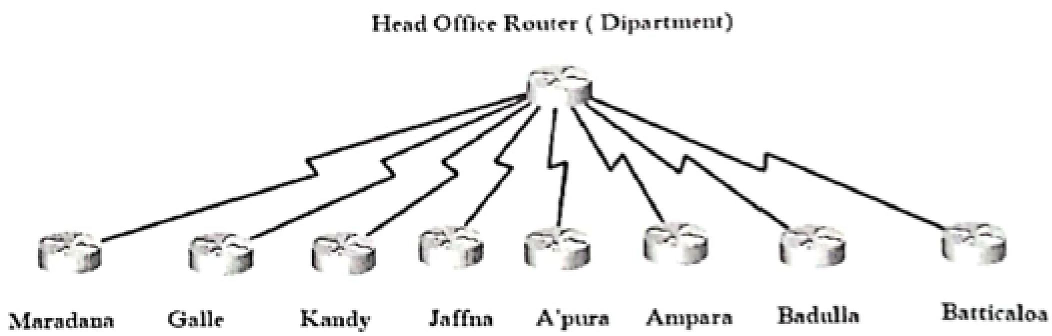
2. The Department of technical education and training planning to interconnect nine College of Technologies. The proposed network diagram as follows.



They have **210.0.0.0/24** IP block and planned to subnet it and use to interconnect above **8 networks**. You are the network administrator and **need to summaries subnet information** as follows.

COT/ Subnet	Network IP	Broadcast IP	Usable IP Range	Sub netmask
Maradana				
Galle				
Kandy				
Jaffna				
A'pura				
Ampara				
Badulla				
Batticaloa				

Copy following table into answer sheet and fill it.

IP Address	Default subnet mask	Class of IP	Public or Private
101.1.1.100	255.0.0.0	Class-A	Public
221.100.100.100			
10.1.1.100			
172.16.150.200			
143.65.100.100			

IP configuration for the server as follows.

IP : 202.110.100.100

Subnet Mask : 255.255.255.192

(a) State the IP given above in CIDR notation.

(b) Write down above IP in binary notation.

(c) What is the network IP for this Network?

(d) What is the broadcast IP for that network?

(e) What is the usable IP rage for this network?

A LAN is configured as follows

Network ID: 172.19.0.0 & subnet mask: 255.255.0.0. Find out followings for LAN.

(a) Network IP Address.

(b) Broadcast IP address.

(c) Usable IP range.

IP configuration for the server as follows:

IP: 200.0.0.100

Subnet Mask: 255.255.255.0 (200.0.0.100/24)

(a) What is the class of this IP?

(b) Is that Public IP or Private IP?

(c) What is the network IP for this Network?

(d) What is the broadcast IP for that network?

(e) What is the usable IP rage for this network?

ABC Company brought 200.0.0.0/24 Network IPv4 address and going to interconnect of their four branches. That means they want four subnets.

Find

(a) Subnet mask for each network.

(b) Network IP for each network?

(c) Usable IP range for each network?

Consider the class C network 192.10.20.0 with subnet mask 255.255.255.224

(a) Find the number of sub networks.

(b) Find the number of host per network.

(c) Find the IP address ranges of 1st subset.

Apex Pvt. Ltd wants design two physically separated networks, each having 13 computers.

The IP addresses of both networks are 20.28.3.0 and 20.28.4.0

a. Find the suitable subnet mask for each of these networks

b. Find network IP address, Host IP addresses and directed broadcasting addresses of each network.

c. Draw a network diagram to communicate with each other and assign IP addresses for all.

Scenario-Based Questions

1. Network Expansion Planning

Your company has been assigned the IP block **192.168.50.0/24**. The IT team needs to create subnets for four different departments: HR, Finance, IT, and Sales. Each department requires at least **50 usable IP addresses**.

- What subnet mask should be used?
- List the network addresses of each subnet.
- What will be the broadcast address of the **Finance** department's subnet?

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2. ISP Address Allocation

An ISP has assigned a **Class B** address **172.25.0.0/16** to a university. The university wants to divide this into **100 subnets** for its departments.

- What subnet mask should the university use?
- How many usable IP addresses will each subnet have?
- What is the network address of the 5th subnet?

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3. Office Network Design

A company has acquired an IP range **200.100.10.0/25** and plans to divide it into two equal subnets for different teams.

- What subnet mask should be used?
- What are the network addresses of both subnets?
- What is the broadcast address of the second subnet?

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4. Network Addressing for a School

A school has **two separate buildings** with **30 computers each**, and each building requires a separate subnet. They have been assigned the IP block **10.10.20.0/24**.

- What subnet mask should they use?
- List the network and broadcast addresses for each building.
- How many unused IPs will be left in each subnet?

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5. Creating Subnets for a Hotel

A hotel needs **5 subnets** from the network **192.168.1.0/24** for different services (Guest WiFi, Staff Network, Management, Security, and IoT devices).

- What subnet mask will allow exactly 5 subnets?
- How many usable hosts will each subnet support?
- List the first subnet's **network address** and **broadcast address**.

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6. Merging Two Networks
A company is merging two branch offices: one uses **192.168.5.0/26**, and the other uses **192.168.5.64/26**. They now want to **combine** these into a single subnet.

- What is the new subnet mask?
- What will be the new **network address**?
- What is the new **broadcast address**?

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7. IP Address Assignments for a Data Center
A data center has **512 servers** and has been given the **192.168.100.0/22** network.

- How many usable IP addresses does this network provide?
- What is the last usable IP address?
- What is the broadcast address of this network?

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8. Internet Cafe IP Configuration
An Internet cafe has **20 computers** and has been allocated **192.168.15.0/27**.

- How many usable hosts are available?
- What is the subnet mask in dotted decimal notation?
- What is the last usable IP address?

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9. Interconnecting Different Branches
A company has **four branches** and has been given the IP **203.0.113.0/24**. They want to divide it into four equal subnets for each branch.

- What subnet mask should they use?
- What are the **network addresses** of each subnet?
- How many usable hosts per subnet will they get?

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10. Network Optimization for an E-commerce Site
An e-commerce company is hosting a **large web application** that needs **1,000 servers** in a single subnet. They are assigned the IP block **198.51.100.0/22**.

- Is this subnet sufficient? If not, what subnet mask should be used?
- How many total IPs does **198.51.100.0/22** provide?
- If they upgrade to a **/21**, how many additional IPs do they gain?

MCQ

1. How many bits are there in an IPv4 address?
A) 16
B) 32
C) 64
D) 128

2. What is the default subnet mask for a Class B network?
A) 255.0.0.0
B) 255.255.0.0
C) 255.255.255.0
D) 255.255.255.255

3. Which of the following is a valid Class A IP address?
A) 192.168.1.1
B) 172.16.0.1
C) 126.50.23.4
D) 224.0.0.1

4. What is the highest possible value of an octet in an IPv4 address?
A) 128
B) 192
C) 255
D) 512

5. How many total addresses are available in a /30 subnet?
A) 4
B) 6
C) 8
D) 2

6. The first octet of a Class C IP address falls within which range?
A) 1 - 126
B) 128 - 191
C) 192 - 223
D) 224 - 239

7. Which of the following IP addresses belongs to Class B?
A) 10.10.10.10
B) 172.20.5.5
C) 192.168.1.1
D) 224.0.0.5

8. What type of IP address is 169.254.1.1?
A) Public
B) Private
C) APIPA (Automatic Private IP Addressing)
D) Multicast

9. Which class of IP address supports the highest number of hosts?
A) Class A
B) Class B
C) Class C
D) Class D

10. Which of the following is a reserved IP address?
A) 8.8.8.8
B) 127.0.0.1
C) 1.1.1.1
D) 192.0.2.1

11. How many usable hosts are available in a /27 subnet?
A) 30
B) 32
C) 62
D) 126

12. What is the broadcast address of the network 192.168.1.0/24?
A) 192.168.1.1
B) 192.168.1.254
C) 192.168.1.255
D) 192.168.0.255

13. What is the subnet mask for a /26 network?
A) 255.255.255.0
B) 255.255.255.192
C) 255.255.255.224
D) 255.255.255.240

14. If you subnet 192.168.10.0/24 into four subnets, what will be the new subnet mask?
A) 255.255.255.128
B) 255.255.255.192
C) 255.255.255.224
D) 255.255.255.240

15. Which of the following subnet masks allows exactly 8 subnets in a Class C network?
A) 255.255.255.128
B) 255.255.255.192
C) 255.255.255.224
D) 255.255.255.240

16. Which of the following is a private IP address?
A) 8.8.8.8
B) 172.16.10.5
C) 192.0.2.5
D) 203.0.113.1

17. Which of the following is a valid public IP address?
A) 192.168.0.1
B) 172.16.255.1

C) 10.0.0.25
D) 150.100.10.10

18. The IP range 192.168.0.0 - 192.168.255.255 belongs to which category?
A) Public
B) Private
C) Multicast
D) Experimental

19. What is the main purpose of a private IP address?
A) To communicate over the internet
B) To prevent conflicts with public IPs
C) To enable multicast traffic
D) To allow anonymous browsing

20. The private IP range for Class A networks is:
A) 172.16.0.0 - 172.31.255.255
B) 10.0.0.0 - 10.255.255.255
C) 192.168.0.0 - 192.168.255.255
D) 100.64.0.0 - 100.127.255.255

21. What is the function of 127.0.0.1?
A) Multicasting
B) Broadcasting
C) Loopback testing
D) Network discovery

22. Which of the following is a multicast address?
A) 192.168.1.1
B) 224.0.0.1
C) 150.100.10.5
D) 10.0.0.1

23. What is the role of a default gateway in a network?
A) Assigns IP addresses dynamically
B) Resolves domain names to IPs
C) Routes traffic between different networks
D) Prevents unauthorized network access

24. Which address is used for limited broadcast traffic?
A) 192.168.1.255
B) 255.255.255.255
C) 224.0.0.1
D) 0.0.0.0

25. Which protocol is used to assign dynamic IP addresses?
A) ARP
B) DHCP
C) ICMP
D) SMTP

26. You configure a network with IP 192.168.1.100/25. What is the network address?
A) 192.168.1.0
B) 192.168.1.64
C) 192.168.1.128
D) 192.168.1.192

27. What does a subnet mask of 255.255.255.252 indicate?
A) Only 2 usable hosts per subnet
B) 30 usable hosts per subnet
C) 62 usable hosts per subnet
D) 126 usable hosts per subnet

28. If a device has an IP address 169.254.x.x, what does it indicate?
A) The device is using a loopback address
B) The device is assigned a private IP
C) The device failed to obtain an IP from DHCP
D) The device is using a public IP

29. If your subnet mask is 255.255.255.224, how many usable host addresses are available?
A) 30
B) 31
C) 32
D) 62

30. What is the primary function of NAT (Network Address Translation)?
A) Assigns static IPs
B) Converts private IPs to public IPs
C) Increases network speed
D) Encrypts IP packets